



Differential Equations: RACE-01

Single Correct Question :

- Find the order and degree of differential equation $\sin^{-1}\left(\frac{dy}{dx}\right) = x + y + 3$
 (A) order : 1 ; degree : not defined (B) order : 1 ; degree : 1
 (C) order : 1 ; degree : 2 (D) None of these
- Find the order and degree of differential equation $\left[\frac{d^2x}{dt^2}\right]^3 + \left[\frac{dx}{dt}\right]^4 - xt = 0$
 (A) order : 3 ; degree : 2 (B) order : 2 ; degree : 4
 (C) order : 2 ; degree : 3 (D) order : 4 ; degree : 2
- Eliminate the arbitrary constants and obtain the differential equation satisfied by $y = 2x + ce^x$
 (A) $y' + y = 2(1 - x)$ (B) $y' - y = 2(1 - x)$
 (C) $y' - y = 2(1 + x)$ (D) $y' + y = 2(1 + x)$
- Eliminate the arbitrary constants and obtain the differential equation satisfied $y = \left(\frac{a}{x^2}\right) + bx$
 (A) $x^2y'' - 2xy' - 2y = 0$ (B) $x^2y'' + 2xy' + 2y = 0$
 (C) $x^2y'' + 2xy' + 2y = 0$ (D) $x^2y'' + 2xy' - 2y = 0$
- Eliminate the arbitrary constants and obtain the differential equation satisfied $y = ae^{2x} + be^{-2x} + c$
 (A) $y''' = 4y'$ (B) $y' = 4y'$ (C) $y'' = 4y'$ (D) $y''' = 2y$
- Obtain the differential equation of the family of circles $x^2 + y^2 + 2gx + 2fy + c = 0$; where g, f & c are arbitrary constants.
 (A) $[1 + (y')^2] \cdot y''' - 3y'(y'')^2 = 0$ (B) $[1 + (y')^2] \cdot y''' + (y'' + 2y')(y'')^2 = 0$
 (C) $[1 + (y')^2] \cdot y''' + 3y'(y'')^2 = 0$ (D) $[1 - (y')^2] \cdot y''' + 3y'(y'')^2 = 0$
- Obtain the differential equation associated with the primitive, $y = c_1 e^{3x} + c_2 e^{2x} + c_3 e^x$, where c_1, c_2, c_3 are arbitrary constants.
 (A) $\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0$ (B) $\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} - 11\frac{dy}{dx} - 6y = 0$
 (C) $\frac{d^3y}{dx^3} + 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} + 6y = 0$ (D) $\frac{d^3y}{dx^3} + 6\frac{d^2y}{dx^2} - 11\frac{dy}{dx} - 6y = 0$
- Solution of differential equation $\sqrt{1 + 4x^2} dy = y^3 x dx$ is
 (A) $\frac{1}{2y^2} = \frac{1}{4}\sqrt{1 + 4x^2} + C$ (B) $\frac{1}{2y^2} = \frac{1}{4}\sqrt{1 - 4x^2} + C$
 (C) $\frac{-1}{2y^2} = \frac{1}{4}\sqrt{1 - 2x^2} + C$ (D) $\frac{-1}{2y^2} = \frac{1}{4}\sqrt{1 + 4x^2} + C$

9. Solution of differential equation $(1 - x^2)(1 - y) dx = xy(1 + y) dy$ is
- (A) $\ln(1 + y)^2 x = c - \frac{1}{2} y^2 + 2y + \frac{1}{2} x^2$ (B) $\ln(1 - y)^2 x = c - \frac{1}{2} y^2 - 2y + \frac{1}{2} x^2$
- (C) $\ln(1 - y)^2 x = c + \frac{1}{2} y^2 + 2y + \frac{1}{2} x^2$ (D) $\ln(1 + y)^2 x = c - \frac{1}{2} y^2 - 2y - \frac{1}{2} x^2$
10. Solution of differential equation $\frac{dy}{dx} + \frac{\sqrt{(x^2 - 1)(y^2 - 1)}}{xy} = 0$ is
- (A) $\sqrt{x^2 - 1} - \sec^{-1} x + \sqrt{y^2 - 1} = c$ (B) $\sqrt{x^2 - 1} + \sec^{-1} x - \sqrt{y^2 - 1} = c$
- (C) $\sqrt{x^2 + 1} - \sec^{-1} x + \sqrt{y^2 + 1} = c$ (D) $\sqrt{x^2 - 1} - \sec^{-1} x - \sqrt{y^2 - 1} = c$

Integer Type Questions :

11. Given $y(0) = 2000$ and $\frac{dy}{dx} = 32000 - 20y^2$, then find the value of $\lim_{x \rightarrow \infty} y(x)$

Answer Key

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	A	C	B	D	A	A	A	D	B	A	40



Differential Equations: RACE-02

Single Correct Questions :

- Solution of the differential equation $\tan^2(x+y)dx - dy = 0$ is
 (A) $2(x+y) = c + \sin 2(x+y)$ (B) $2(x-y) = c + \sin 2(x+y)$
 (C) $2(x-y) = c + \sin 2(x-y)$ (D) $2(x+y) = c - \sin 2(x+y)$
- Solution of the differential equation $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$ is
 (A) $\tan\left(\frac{x+y}{2}\right) = ae^x - 1$ (B) $\tan\left(\frac{x-y}{2}\right) = ae^x - 1$
 (C) $\tan\left(\frac{x+y}{2}\right) = ae^{-x} - 1$ (D) $\tan\left(\frac{x+y}{2}\right) = ae^x + 1$
- Solution of the differential equation $\sin^{-1}\left(\frac{dy}{dx}\right) = x+y$ is
 (A) $\tan(x+y) - \sec(x+y) = x+c$ (B) $\tan(x+y) + \sec(x+y) = x+c$
 (C) $\tan(x+y) - \sec(x+y) + x = c$ (D) $\tan(x+y) + \sec(x+y) + x = c$
- Solution of the differential equation $\frac{dy}{dx} = \frac{x-2y+5}{2x+y-1}$ is
 (A) $x^2 + 4xy - y^2 - 10x + 2y = c'$ (B) $x^2 - 4xy + y^2 + 10x - 2y = c'$
 (C) $x^2 - 4xy - y^2 + 10x + 2y = c'$ (D) $x^2 + 4xy + y^2 - 10x - 2y = c'$
- Solution of the differential equation $\frac{dy}{dx} = \frac{2x-y+3}{x+2y+4}$ is
 (A) $y^2 + x^2 - xy + 4y - 3x + c = 0$ (B) $y^2 - x^2 + xy - 4y + 3x + c = 0$
 (C) $y^2 - x^2 + xy + 4y - 3x + c = 0$ (D) $y^2 + x^2 + xy - 4y + 3x + c = 0$
- Solution of the differential equation $(y - xy^2)dx - (x + x^2y)dy = 0$ is
 (A) $x = cye^{-xy}$ (B) $y = cxe^{xy}$ (C) $x^2 = cye^{xy}$ (D) $x = cye^{xy}$
- Solution of the differential equation $ydx - xdy = xydy - x^2dx$ is
 (A) $\sqrt{x^2 - y^2} = \sin^{-1}\left(\frac{y}{x}\right) + c$ (B) $\sqrt{x^2 + y^2} = \sin^{-1}\left(\frac{y}{x}\right) + c$
 (C) $\sqrt{x^2 - y^2} = \sin^{-1}\left(\frac{x}{y}\right) + c$ (D) $\sqrt{x^2 + y^2} = -\sin^{-1}\left(\frac{y}{x}\right) + c$
- Solution of the differential equation $x dx + y dy = x(x dy - y dx)$ is
 (A) $(y-1)^2 = c(x^2 + y^2)$ (B) $(y+1)^2 = c(x^2 + y^2)$
 (C) $(y+1)^2 = c(x^2 - y^2)$ (D) $(y-1)^2 = c(x^2 - y^2)$

Answer Key

Question	1	2	3	4	5	6	7	8
Answer	B	A	A	C	C	D	A	B



Differential Equations: RACE-03

Single Correct Questions :

- Solution of the differential equation $y' = \frac{3x-y}{x+y}$ is

(A) $(3x+y)(x-y) = c$ (B) $(3x+y)(x+y) = c$
 (C) $(3x-y)(x-y) = c$ (D) $(x+3y)(x-y) = c$
- Solution of the differential equation $(x - y \ln y + y \ln x) dx + x(\ln y - \ln x) dy = 0$ is

(A) $y \ln \left(\frac{y}{x} \right) + y + x \ln x + cx = 0$ (B) $y \ln \left(\frac{y}{x} \right) - y + x \ln x + cx = 0$
 (C) $y \ln \left(\frac{y}{x} \right) - y - x \ln x + cx = 0$ (D) $y \ln \left(\frac{x}{y} \right) - y + x \ln x + cx = 0$
- Solution of the differential equation $(3xy + y^2)dx + (x^2 + xy)dy = 0, y(1) = 1$ is

(A) $x^2y(2x - y) = 3$ (B) $x^2y(x + 2y) = 3$
 (C) $x^2y(2x + y) = 3$ (D) $x^2y(2x + y) = 0$
- Solution of the differential equation $\frac{dy}{dx} = \frac{x+2y-5}{2x+y-4}$ is

(A) $x - y - 3 = C(x - y + 1)^3$ (B) $x + y - 3 = C(x + y + 1)^3$
 (C) $x + y - 5 = C(x - y + 1)^3$ (D) $x + y - 3 = C(x - y + 1)^3$
- Solution of the differential equation $\frac{dy}{dx} = \frac{x+y+1}{2x+2y+3}$ is

(A) $x + y + \frac{4}{3} = ce^{4(x-2y)}$ (B) $x + y + \frac{4}{3} = ce^{3(x+2y)}$
 (C) $x + y + \frac{4}{3} = ce^{3(x-2y)}$ (D) $x - y + \frac{4}{3} = ce^{3(x-2y)}$
- Solution of the differential equation $(x - y)dy = (x + y + 1)dx$ is

(A) $\arctan \frac{2y+1}{2x+1} = \ln c \sqrt{x^2 - y^2 + x + y + \frac{1}{2}}$
 (B) $\arctan \frac{2y+1}{2x+1} = \ln c \sqrt{x^2 + y^2 + x + y + \frac{1}{2}}$
 (C) $\arctan \frac{2x+1}{2y+1} = \ln c \sqrt{x^2 + y^2 + x + y + \frac{1}{2}}$
 (D) $\arctan \frac{2y+1}{2x+1} = \ln c \sqrt{x^2 + y^2 + x - y - \frac{1}{2}}$

7. Solution of the differential equation $(x - a) \frac{dy}{dx} + 3y = 12(x - a)^3, x > a > 0$ is

(A) $y = 2(x - a)^3 + \frac{c}{(x + a)^3}$

(B) $y = 2(x + a)^3 + \frac{c}{(x - a)^3}$

(C) $y = 2(x - a)^3 + \frac{c}{(x - a)^3}$

(D) $y = 3(x - a)^3 + \frac{c}{(x - a)^3}$

8. Solution of the differential equation $y \ln y \, dx + (x - \ln y) \, dy = 0$ is

(A) $2x \ln y = \ln^2 x + C$

(B) $2x \ln y = \ln^2 y + C$

(C) $2y \ln y = \ln^2 y + C$

(D) $4x \ln y = \ln^2 y + C$

Answer Key

Question	1	2	3	4	5	6	7	8
Answer	A	B	C	D	C	B	C	B



Differential Equations: RACE-04

Single Correct Questions :

1. $x(x-1) \frac{dy}{dx} - (x-2)y = x^3(2x-1)$

(A) $y(x-1) = x^2(x^2 - x + c)$

(B) $y(x+1) = x^2(x^2 - x + c)$

(C) $y(x-1) = x^2(x^2 + x + c)$

(D) $y(x+1) = x^2(x^2 + x + c)$

2. $xdy - \{y + xy^3(1 + \ln x)\}dx = 0$

(A) $\frac{x^2}{y^2} = \frac{2}{3}x^3\left(\frac{2}{3} + \ln x\right) + c$

(B) $\frac{x^2}{y^2} = -\frac{2}{3}x^3\left(\frac{2}{3} + \ln x\right) + c$

(C) $\frac{x^2}{y^2} = -\frac{2}{3}x^3\left(\frac{1}{3} + \ln x\right) + c$

(D) $\frac{x^2}{y^2} = -\frac{3}{2}x^3\left(\frac{2}{3} + \ln x\right) + c$

3. $x^2y - x^3 \frac{dy}{dx} = y^4 \cos x$

(A) $x^3y^{-3} = 3\cos x + c$

(B) $x^{-3}y^3 = 3\sin x + c$

(C) $x^3y^3 = 3\sin x + c$

(D) $x^3y^{-3} = 3\sin x + c$

4. $x(x^2 + 1) \frac{dy}{dx} = y(1 - x^2) + x^3 \cdot \ln x$

(A) $4(x^2 + 1)y - x^3(1 - 2 \ln x) = cx$

(B) $4(x^2 + 1)y + x^3(1 + 2 \ln x) = cx$

(C) $4(x^2 + 1)y + x^3(1 - 2 \ln x) = cx$

(D) $(x^2 + 1)y + x^3(1 - 2 \ln x) = cx$

5. $\left(\frac{dy}{dx}\right)^2 - (x+y) \frac{dy}{dx} + xy = 0$

(A) $y = ce^x; y = c + \frac{x^2}{2}$

(B) $y = ce^x; y = c - \frac{x^2}{2}$

(C) $x = ce^y; y = c + \frac{x^2}{2}$

(D) $y = ce^x; y = c + \frac{x^2}{4}$

6. $\frac{dy}{dx} = e^{x-y}(e^x - e^y)$

(A) $e^y = c \cdot \exp(-e^x) + e^x - 1$

(B) $e^y = c \cdot \exp(-e^x) + e^x + 1$

(C) $e^y = c \cdot \exp(e^x) + e^x - 1$

(D) $e^y = c \cdot \exp(e^x) - e^x - 1$

7. $x dx + y dy + 4y^3(x^2 + y^2)dy = 0.$

(A) $\ln(x^2 + y^2) + y^4 = C$

(B) $1/3 \ln(x^2 + y^2) + y^4 = C$

(C) $1/2 \ln(x^2 + y^2) + y^4 = C$

(D) $1/2 \ln(x^2 - y^2) + y^4 = C$

Integer Type Questions :

8. Let f be a differentiable function such that $x^2 f(x) - x = 4 \int_0^x t f(t) dt$, $f(1) = \frac{2}{3}$.

Then $18 f(3) = \underline{\hspace{2cm}}$

9. If $y = y(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{4x}{(x^2-1)} y = \frac{x+2}{(x^2-1)^{\frac{5}{2}}}$, $x > 1$ such that

$y(2) = \frac{2}{9} \log_e(2 + \sqrt{3})$ and $y(\sqrt{2}) = \alpha \log_e(\sqrt{\alpha} + \beta) + \beta - \sqrt{\gamma}$, $\alpha, \beta, \gamma \in N$, then $\alpha\beta\gamma$ is equal to ____.

Answer Key

Question	1	2	3	4	5	6	7	8	9
Answer	A	B	D	C	A	A	C	160	6